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RICE DDGS

Distillers Dried Grain Solubles

TECHNICAL BENEFITS

- DDGS has a high and favorable amino acid profile.
- Phytic acid is controlled in our DDGS, making it more digestible.
- Low moisture content in DDGS prevents the growth of harmful microorganisms, increasing shelf life.
- Antinutritional factors are kept low in DDGS.
- Our distillery process is sulfuric acid free, eliminating nutritional problems in ruminant diets.
- Fermentation of DDGS provides vitamin B-complex, promotes palatability, and processes feed consumption through 3-5% yeast cells.



Rice DDGS has a high nutritional value, providing 45% protein & vitamin B-Complex. This helps to increase the feed consumption while reducing the cost of feed, leading to optimum yields. With Rice DDGS, you can be sure that your livestock is getting the nutrition they need. You can view its nutritional facts below.

NUTRITION FACTOR

Parameters	Normal	Hi Pro
Protein	45-50%	50-55%
Moisture % (Max)	8-11%	8-11%
Fat %	2-3%	2-3%
Fiber %	5-6%	5-6%
Sand Silica % (Max)	1%	1%
M Energy (Kcal / KG)	3000- 2000	3000- 2000

Providing

45 %- 50% protein

With highly digestible amino acid

Application

- Swine Feed
- Cattle feed manufacturing units
- Poultry feed manufacturing units
- Aquaculture feed manufacturing units

AMINO ACID PROFILE

Parameters	Value
Arginine	3.09
Histidine	-
Isoleucine	2.45
Leucine	4.4
Lysine	2.36
Methionine	1.23
Phenylalanine	2.94
Threonine	2.7
Tryptophan	0.6
Valine	2.33
Alanine	3.94
Aspartic Acid	4.92
Cystine	1.33
Glutamic Acid	4.84
Glycine	2.68
Proline	2.66
Serine	2.63
Tyrosine	-

CORN DDGS

Distillers Dried Grain Solubles

TECHNICAL BENEFITS

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- Fermentation of DDGS provides vitamin B-complexes, promotes palatability, and increases feed consumption through 3-5% yeast cells.



Corn DDGS has a high nutritional value, providing 38% profat. This helps to to increase the feed consumption while reducing the cost of feed.

NUTRITION FACTOR

Parameters	
Protein	26% - 30%
Moisture % (Max)	8% - 11%
Fat %	8% - 10%
Fiber %	5% - 6%
Sand Silica % (Max)	1%
Aflatoxin	30 PPB (Approx)
M Energy (Kcal / KG)	2800- 3000

Providing

36% - 40% Profat

With highly digestible amino acid

Corn DDGS has a high nutritional value, providing 38% profat. This helps to to increase the feed consumption while reducing the cost of feed.

Application

- Cattle feed manufacturing unit
- Aqua feed manufacturing unit

AMINO ACID PROFILE

Parameters	
ADF	10.00%
NDF	20.00%
Ash	5.00%
Hemicellulose (DM)	21.00%
Xylose	6.00%
Metabolic Energy (ME)	3384 kcal/kg
DE	3946 kcl/kg
Calcium	0.05%
Phosphorus	0.82%
Sodium	0.20%
Potassium	1.15%
Magnesium	0.29%
TDNRumen (RUP)	74
TDNPoultry/Aqua	85
Sulfur	0.85%
Copper, ppm	4.00
Iron, ppm	70.00
Manganese, ppm	22.00
Zinc, ppm	51.00
Lysine	1.00%
Methionine	0.55%

RAPESEED MEAL

TECHNICAL BENEFITS

Rapeseed meal is a protein-rich feed for livestock, with good amino acid profile and significant methionine content. It's high in fibre and lignin, limiting its use in monogastric diets. It's commonly used as a substitute for soybean meal and is highly palatable for ruminants. Its residual oil content varies depending on the processing method.



Applications of Rapeseed Meal:

- Cattle Feed – Used in dairy and beef cattle rations as a protein supplement.
- Poultry Feed – Included in chicken and layer feed formulations to support growth and egg production.
- Aquaculture Feed – Used in fish and shrimp feed as an economical plant-based protein source.

NUTRITION FACTOR

Main Analysis	Unit	Avg
Dry matter	% as fed	89
Crude protein	% DM	38.1
Crude fibre	% DM	14.3
Neutral detergent fibre	% DM	31.6
Acid detergent fibre	% DM	20.7
Lignin	% DM	9.7
Ether extract	% DM	2.4
Ash	% DM	7.6
Insoluble ash	% DM	0.3
Starch (polarimetry)	% DM	6.3
Starch (enzymatic)	% DM	1.6
Total sugars	% DM	10.5
Gross energy	MJ/ kg DM	19.3

CORN GLUTEN MEAL

DESCRIPTION

Corn gluten meal is a by-product of the manufacture of maize starch (and glucose syrup) by the wet-milling process. Corn gluten meal is a protein-rich feed, containing some 65% crude protein (DM), used as a source of protein, energy and pigments for livestock species making diets. It is also valued in pet food for its high protein digestibility. Corn gluten meal is also used as a fertilizer and pre-emergent weed killer.

Corn gluten meal is produced wherever maize is used for starch production. Its production has become relatively common safely enhanced it also when produced by dry milling, which yields main products rather than corn gluten meal and corn gluten feed.



NUTRITION FACTOR

Parameters	Value in %age
Protein %	60-70
Moisture % (Max)	10
Fat %	4-8
Sand Silica % (Max)	2
M Energy (KCal / KG)	4131
Pepsin Digestibility	85-90

- Corn Gluten Meal is a protein-rich by product of maize starch manufacture by wet-milling process
- Contains about 65% crude protein (DM) and minimal amounts of starch and fiber
- Used as a source of protein, energy, and pigments for livestock, including fish, and in pet food for high protein digestibility
- Also used as a fertilizer and pre-emergent weed fiber
- Corn gluten meal production has become relatively constant as ethanol is mainly produced by dry milling, yielding corn distillers instead

Table: Composition of amino acids in Corn Gluten Feed/Maize Fiber/CoFeed

Parameters	Value in %age
Arginine (%)	2.11
Histidine (%)	1.42
Isoleucine (%)	3.01
Leucine (%)	12.43
Lysine (%)	1.66
Methionine (%)	3.65
Phenylalanine (%)	3.65
Threonine (%)	1.99
Tryptophan (%)	6.3
Valine (%)	3.39
Cystine (%)	1.08
TSAA Met+Cys (%)	3.76
Tyrosine (%)	6.76
Phe+Tyr (%)	7.41
Glutamic (%)	8.71
Aspartic (%)	3.24
Glycine (%)	3.23
Valine (%)	3.23
Alanine (%)	8.8
Sum EAAs (%)	20.71
Sum NEAAs (%)	30.54
Taurine (%)	0

CORN GLUTEN FEED

DESCRIPTION

- Corn gluten feed is a byproduct of maize grain wet-milling for starch or ethanol production.
- It is primarily primarily of bran bran and steep liquor, and may even contain distillers grains, germ meat and cracked corn screenings.
- Corn gluten feed is a moderately high source of protein (20-25% DM), and is commonly used in ruminant diets.
- Its nutritional attributes, including energy and protein content, vary depending on the milling process and the proportion of steep liquor in the diets.



NUTRITION FACTOR

Parameters	Value in %age
Protein %	18-20
Moisture % (Max)	10
Fat %	3
Sand Silica % (Max)	3
M Energy (KCal / KG)	2000

COMMON NAMES

- Corn gluten feed is a byproduct of maize wet-milling and consists of maize bran and steep liquor. It can be dry or wet and its composition varies widely. It's a major food ingredient for ruminants, with moderate protein content and high in cell wall constituents.

NUTRITIONAL ATTRIBUTES

- Corn gluten feed is a major feed ingredient for beef and dairy cattle.
- It is available as wet (40-60% DM) or dry (88% DM) product.
- Contains 20-25% DM protein, less than corn gluten meal and most oil meals but more than cereal grains and milling by-products.
- Richer in cell wall components to sells from 6-10% DM, NDF 31-45% DM, ADF 6-13% DM) than maize grain which limits to use in pig and poultry diets.
- Contains under 4% DM crude far and variable amounts of residual starch (11-30% DM).
- Ash content is about 7% DM.

Table:

- Composition of amino acids in Corn Gluten Feed/Maize Fiber/CoFeed

Parameters	Value in %age
Lysine	0.67
Threonine	0.72
Methionine	0.35
Cystine	0.39
Methionine + Cystine	0.34
Tryptophan	0.14
Isoleucine	0.77
Valine	0.36
Leucine	1.17
Phenylalanine	0.99
Tyrosine	0.98
Phenylalanine + Tyrosine	1.29
Histidine	0.88
Arginine	1.20
Alanine	1.32
Aspartic Acid	1.21
Glutamic Acid	4.84
Glycine	1.69
Proline	2.66
Serine	2.63

OTHER PRODUCTS

SOYABEAN MEAL

- Soybean meal is a by-product of extracting oil from whole soybeans.
- It is high in protein and energy, making it a popular protein supplement in North America.
- Soybean meal is a palatable feedstuff and can be used as the major protein supplement in dairy cattle rations.



No.	Nutrients	Results
1	Protein	46.5%
2	Urease / Protease	0.20 (pH Change / 1 hr)
3	Moisture	9.4%
4	Total Fat	1.2%
5	Total Carbohydrate	33.4%
6	Total Ash	6.2%
7	Fiber%	4.8%

Typical Analysis (Solvent extracted soybean meal 46%)	
Crude Protein	46.5%
Crude Fat	1.0%
Crude Fiber	3.5%
Total Carbohydrate (NFE)	33.0%
Acid Detergent Fiber	6.8%
Calcium	0.3%
Phosphorus	0.7%
Total Digestible Nutrients	80.0%
Net energy in lactation	1.53 Meal / kg (DM)

DEOILED RICE BRAN

- De-oiled rice bran (DORB) is a by-product of rice bran oil extraction process.
- DORB is commonly used in animal feed for cattle, poultry, fish, and other species due to its moderate protein content and economical price.
- DORB is also utilized in the production of other products such as boiler fuel, silica gel, and insulating bricks.



Parameters	Unit	Value (Approx.)
Dry Matter	% DM	88.0
Crude Protein	% DM	12.0
Crude Fat	% DM	14.0
Crude Fiber	% DM	11.0
Neutral Detergent Fiber	% DM	38.0
Crude Protein	% DM	21.0
Lignin	% DM	8.0
Ether Extract	% DM	12.0

TURMERIC OIL

- The active substances in the Turmeric oil are Curcuminoids. Curcuminoids have a wide spectrum of bioactive properties including antioxidant, anticancer, antifungal, antiarthrototoxic antiviral, antiprotazoal and anti-inflammatory properties



Parameters	Turmeric Oil
Curcumin	Min 18%
FFA	3-5%
M Energy (Kcal/kg)	9000- 10000

Feed Color

Feed colour can be significantly improved with Turmeric oil @ 1-2 Kg/ton of feed.

CORN STARCH

- Corn starch is a white, tasteless, odorless powder.
- It is used in food processing, papermaking, and the production of industrial adhesives.
- Corn starch is also used as a component of many cosmetics and old pharmaceutical products.
- It has been used as a lubricant in surgical gloves.



Test Parameters	Specifications	Results
Description	A white, odorless powder.	Absent
Moisture	Not more than 13.0%	8.22
Total Ash	Not more than 0.20%	0.12
pH (5% Starch Solution)	4.0 - 7.0	5.31
Protein (N×6.25)	Not more than 0.50%	0.25
Starch Content	Not less than 99.0%	99.85

ANIMAL BASED PROTEIN

FISH MEAL

- Fishmeal is made by cooking, pressing, drying, and grinding fish or fish waste.
- It is a premium source of animal feed for poultry, swine, mink, farm-raised fish, and pets.
- Fish meal is high in protein concentrated and a good source of calcium, phosphorus, and other minerals.
- It is particularly popular in aquaculture feeds due to its high protein content and excellent amino acid balance for aquatic animals.



Parameters	Specifications
Name of Product	Steam Dried Fish Meal
Protein	60 ± 5%
Moisture	10 %
TVN	120 mg/100 gm
Fat	7 ± 2%
E. Coli	Nil
Salmonella	Absent/25 gm
Shigella	Absent/25 gm

POULTRY MEAL

- Poultry meal consists of the milled, rendered, and cleaned parts of slaughtered poultry.
- It is an excellent source of protein and used extensively in companion animal diets.
- Improvements in digestibility have enhanced its usage in other animal species, such as in aquaculture and starting pig diets.
- Broiler chickens, laying hens, and turkeys are the major sources of poultry raw materials.



Parameters	Specification
Name of Product	Poultry Meal
Crude Protein	62.00%
Moisture	5.00%
Ether Extract	20.50%
Crude Fiber	1.50%